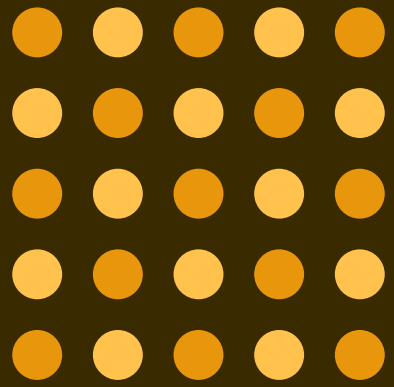
UNISO · Introduction to Artificial Intelligence



Data — The foundation of AI

Every layer is hungry. What we feed it decides everything it becomes.

Wednesday 1:15–3:40 PM · Data —The foundation of AI



Two clinic apps claim to give health advice in Somali. App A was trained on 2,000 Somali patient records. App B is a clever demo with... no data. Which app would you trust — and what would you need to SEE before deciding?

We return to this question at the end of the session.

By the end of this session you can

By the end of this session you can…

- Name the 5 main types of data with one example each
- Explain structured vs unstructured data
- Say why 'garbage in, garbage out' is the first law of AI
- Spot 2 ways bias sneaks into data
- Build a clean dataset in Google Sheets — and judge if it is good enough to train on

Five types of data

Numbers

- prices, rainfall mm
- exam marks

¶ Text

- messages, reviews
- news, tweets

◉ Images

- crop photos, faces
- receipts

♪ Audio

- voice notes, calls
- radio news

▶ Video

- market footage
- class recordings

▦ Logs & streams

- sensors, CCTV
- device records

Structured = tidy tables (rows + columns). Unstructured = messy human stuff — AI's big trick is taming the messy.

Where data comes from

People

- surveys, WhatsApp, reviews
- you are a data source

Devices

- phones, sensors, cameras
- weather stations

Hospitals

- patient records
- test results

Markets & shops

- prices, sales, livestock
- customer behaviour

Satellites

- rainfall, land use
- city growth

Organisations

- ministries, banks
- UN/FAO public data

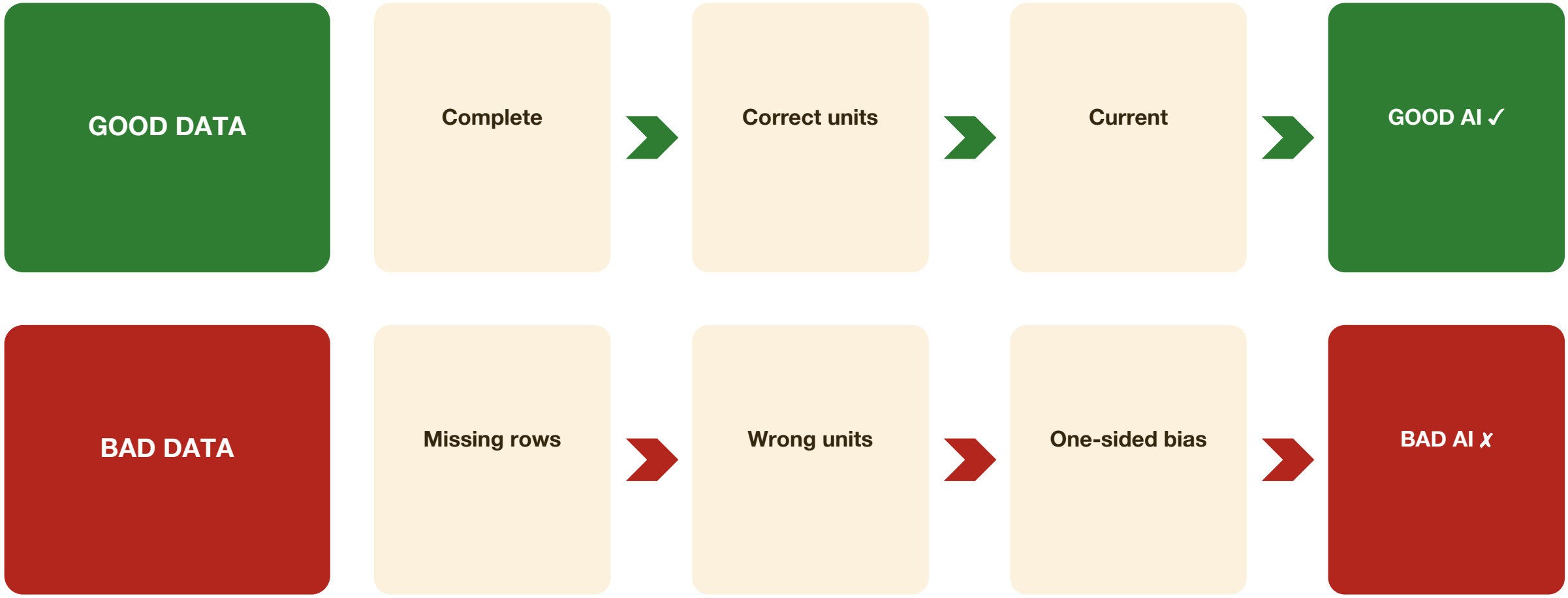
Most AI data is collected by accident — then organised by someone.

What makes data good (or bad)

Quality	What to check	Trap example
Complete	No missing rows or cells	August has no rainfall number
Correct	Right numbers, right units	Inches in a millimetre column
Consistent	Same format everywhere	'November' written two ways
Current	Recent enough for the question	Last year's prices used today
Ethical	Collected with permission	Private patient data shared

Your five-word checklist — used in every project from W11 to W12's evaluation.

Garbage in, garbage out



Garbage In, Garbage Out — the first law of AI. The model does only what the data allows.

Bias — when data is one-sided

ANALOGY — think of it like…

A market survey done only at the university gate will describe only students — not the city. Ask only at the gate, and your 'city view' is secretly a campus view.

FORMAL DEFINITION

Bias in data means the data does not fairly represent the world it claims to describe — some people, places, or cases are missing or over-represented.

SOMALI CONTEXT — halka aan ka helno

An AI trained mostly on English internet text understands Somali poorly. The data was not wrong — it was just gathered without Mogadishu in mind.

Live Demo — thin data, confident wrong answer

YOU TYPE

Ask: 'How much does a sack of rice cost in Mogadishu today?'

Then: 'What was the rainfall in Mogadishu in November 2023?'

AI RETURNS

(Expect a confident but outdated or vague answer. Ask 'where did you learn that?' — the AI will admit its data window and sources.)

Takeaway: Confident ≠ correct. Always ask the assistant 'where did you learn that?' — that is data checking.

Data mistakes students make

The mistake	Why it happens / the fix
'More data is always better'	More WRONG data is worse than less good data. Quality beats quantity.
'Data is neutral — numbers don't lie'	Numbers are collected by people, so they carry people's blind spots.
'One clean sheet is enough'	One season, one market, one school — small data claims big things at its peril.
'I can use anyone's data'	Using personal data needs permission. Ethics is part of data work from Week 3.

Lab Time — Mogadishu rainfall dataset

OBJECTIVE

Build a structured, clean, charted dataset of monthly rainfall in Google Sheets — then judge whether it is enough to advise a farmer.

STEPS

- 1. Create the sheet: columns Month, Rainfall (mm), Season, Notes
- 2. Enter 12 months of rainfall (data sheet provided — type it exactly)
- 3. Clean it: find missing cells, fix units, check consistency — log every fix in Notes
- 4. Add season labels: Gu / Xagaa / Dayr / Jiilaal, then make a line chart

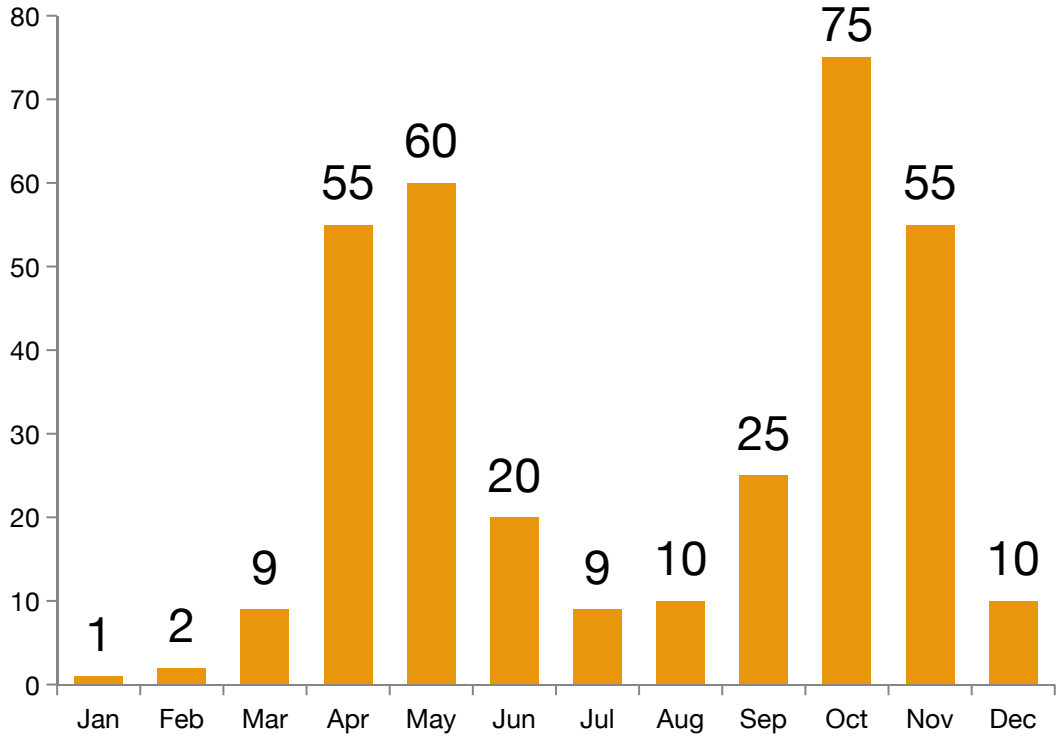
DELIVERABLE — Clean sheet + chart + one-sentence answer: is this enough data to advise a farmer?

Lab checkpoint

STOP — answer before continuing

Your dataset had an 'inches' cell and a missing August. Both got fixed. Which one hurt the final chart more — and why?

Tell your buddy. Then show the instructor.



Independent challenge

SOLO · NO HELP · 10 MIN

- Find one 'official-looking' number in your life — a price, a statistic, an average
- Interrogate it: who collected it? when? from whom? does it include your area?
- Write one sentence: is it trustworthy enough to act on?
- Share your verdict with the class

Done? Show the instructor. Explain it aloud — that is the check.

Quick Quiz — Week 3

5 questions · 15 minutes · answers stay with the instructor

Q1 — Name two types of data and one example of each.

Q2 — A table with rows and columns is which kind of data?

Q3 — What does 'garbage in, garbage out' mean in one sentence?

Q4 — A weather model trained only on two dry years predicts drought every year. What is the real problem?

Q5 — True or false: a dataset is trustworthy because it contains numbers. Explain.

Your project

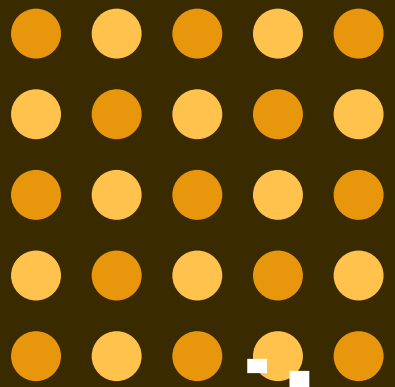
THIS WEEK FEEDS YOUR GROUP PROJECT

- Every project in the bank needs data — your project's quality starts here
- Tonight's homework: collect your own 15-row table (any countable thing)
- That table becomes the data core of your Week 11 plan and your Week 12 evaluation
- Week 4 splits your data into TRAINING and TESTING — the test half is the honesty check

Project pipeline: choose a problem (W11) → build (W11–W12) → test (W12) → document (W13) → defend (W14).

Today's takeaways

- Data is the fuel — no data, no AI
- Structured (tables) vs unstructured (messy) — AI tames the messy
- GIGO: garbage in, garbage out — the first law of AI
- Good data: complete, correct, consistent, current, ethical
- Bias = missing people; check WHO is inside the data



Own it.

Feed the machine well, and it will earn its keep.

Next week: How a machine learns — algorithms, models, training, and your own 15 rows of data.